Article presentation : Shape deformation analysis from the optimal control viewpoint - MVA 2025/2026

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April 13, 2026

1 Introduction

The aim of this project is to read and present an article related to the MVA course 'Geometry and Spaces of Forms'. This report contains a selection of proofs and theorems from the paper: "Shape deformation analysis from the optimal control viewpoint" [1], published in the Journal de Mathématiques Pures et Appliquées in 2015, which presents a fairly general framework for form spaces.

In addition, we will briefly present the following article "Structured Deformation Modeling with Implicit Deformation Modules" and "A sub-Riemannian modular framework for diffeomorphism based analysis of shape ensembles" [2, 3] , which develops an extension of the framework of form spaces discussed in class, in a situation where the local metric on the form space is not induced by a right-invariant metric on the group of diffeomorphisms.

This report provide a concise **synthesis** of the three papers, focusing on their central definitions and main theoretical results.

2 Modelling shape deformation problems with optimal control

Let X and Y be Banach spaces, M be an open subset of X. For every $l \in \mathbb{N}^\star$, $\mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d)$ is the space of vector fields of class $\mathcal{C}^l$ whose derivatives of order less than or equal to l converge to zero at infinity.And $\mathrm{Diff}_0^l(\mathbb{R}^d)$ is the set of all diffeomorphisms of class $\mathcal{C}^l$ that converge to identity at infinity. We use $\|\cdot\|_l$ the usual norm on $\mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d)$ turning it into a Banach space.

2.1 Deformations and RKHS of vector fields

Definition 1. With $v \in L^1([0,1]; \mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d))$ a velocity field and $\varphi \in W^{1,1}([0,1]; \mathrm{Diff}_0^l(\mathbb{R}^d))$ the associated flow, also called *deformation* of class $\mathcal{C}^l$.

$$\forall t \in [0,1],\ \dot{\varphi}(t) = v(t) \circ \varphi(t), \quad \varphi(0) = \mathrm{Id}_{\mathbb{R}^d} \tag{1}$$

Proposition 1. (Arguillere et al. [1]) The set of deformations of $\mathbb{R}^d$ of class $\mathcal{C}^l$ is exactly the set:

$$\left\{ \varphi \in W^{1,1}([0,1]; \mathrm{Diff}_0^l(\mathbb{R}^d)) \,|\, \varphi(0) = \mathrm{Id}_{\mathbb{R}^d} \right\}$$

And there exists $(D_i)_{i\in\mathbb{N}}$ such that for every $i \in \{1, \cdots, l\}$

$$\sup_{t\in[0,1]} \|\varphi(t) - \mathrm{Id}_{\mathbb{R}^d}\|_i \leq D_i \exp\left(D_i \int_0^1 \|v(s)\|_i ds \right) \tag{2}$$

Definition 2. A *Reproducing Kernel Vector Space* of vector fields of class $\mathcal{C}^l$ is a Hilbert space $(V, \langle\cdot,\cdot\rangle_V)$ of vector fields on $\mathbb{R}^d$, such that $V \subset C_0^l(\mathbb{R}^d, \mathbb{R}^d)$ with continuous inclusion (i.e. $\exists C > 0$, $\|v\|_l \leq C\|v\|_V$).

Let $(V, \langle\cdot,\cdot\rangle_V)$ be an Hilbert space. Let $K_V : V^\star \to V$ be the inverse of the Riesz bijective isometry. For all $x, y, b \in \mathbb{R}^d$ let $\delta_y^b \in V^\star$ such that $(\delta_y^b, v)_{V^\star,V} = \langle v(y), b\rangle_{\mathbb{R}^d}$.The *reproducing kernel* K of V is then defined as $K(x,y)b = K_V\delta_y^b(x)$ i.e $K_V\delta_y^b = K(\cdot,y)b$. Finally, we have the *reproducing property* $\langle K(\cdot,x)a, K(\cdot,y)b\rangle = a^T K(x,y)b$

2.2 From shape space problems to optimal control

Definition 3. An *action* $(\varphi, q) \in \mathrm{Diff}_0^l(\mathbb{R}^d) \times M \mapsto \varphi \cdot q \in M$ is such that $\mathrm{Id}_{\mathbb{R}^d} \cdot q = q$ and $\varphi \cdot (\psi \cdot q) = (\varphi \circ \psi) \cdot q$ for all $\psi, \varphi \in \mathrm{Diff}_0^l(\mathbb{R}^d)$ and $q \in M$.

Definition 4. The *infinitesimal action* ξ_q of $\mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d)$ is defined as the differential of $\varphi \mapsto \varphi \cdot q$ at identity.

Furthermore, the action needs to be *compatible with the properties of the group structure* of $\mathrm{Diff}_0^l(\mathbb{R}^d)$ i.e. differentiable at identity, Lipschitz with respect to $\| \cdot \|_l$, and some regularity of $(q, v) \mapsto \xi_q v$.

Proposition 2. (Arguillere et al. [1]) For every $q \in M$, $\varphi \mapsto \varphi \cdot q$ is $\mathcal{C}^1$ and its differential at φ is $\delta\varphi \mapsto \xi_{\varphi \cdot q} \delta\varphi \circ \varphi^{-1}$

As a consequence, with $q(t) = \varphi(t) \cdot q_0$ then $\dot{q}(t) = \xi_{q(t)} v(t)$ with $v(t) = \dot{\varphi}(t) \circ \varphi(t)^{-1}$.

Problem 1. Let $q_0 \in M$, and let $C : (q, v) \in M \times V \mapsto C_q v \in Y$ linear with respect to v. Let $g : M \to \mathbb{R}$ be a function. We consider the problem of minimizing the functional:

$$J_1(q, v) = \frac{1}{2} \int_0^1 \|v(t)\|_V^2 dt + g(q(1)) \tag{3}$$

Over all $q(\cdot) \in W_{q_0}^{1,1}([0, 1]; M)$, $v(\cdot) \in L^2([0, 1], V)$ such that $\dot{q}(t) = \xi_{q(t)} v(t)$ and $C_{q(t)} v(t) = 0$ for almost every $t \in [0, 1]$.

Definition 5. A state $q \in M$ of order l is said to have *compact support* if for some compact subset U of $\mathbb{R}^d$, for some $\gamma > 0$.

$$\forall \varphi_1, \varphi_2, \quad \|\varphi_1 \cdot q - \varphi_2 \cdot q\| \leq \gamma \|(\varphi_1 - \varphi_2)_{|U}\|_l$$

Theorem 1. (Arguillere et al. [1]) Assume that V is an RKHS of vector fields of class C^{l+1} on $\mathbb{R}^d$, that $q \mapsto C_q$ is continuous, and that g is bounded below and lower semi-continuous. If q_0 has compact support, then problem 1 has at least one solution.

Definition 6. The *induced kernel operator* at q is defined as $K_q = \xi_q K_V \xi_q^\star$. It is a continuous operator $K_q : X^\star \to X$.

There exist an unique $u \in X^\star$ such that $\xi_q v = K_q u$. And $v' = K_V \xi_q^\star u$ achieves the minimal norm such that $\xi_q v = \xi_q v'$. Indeed if $\mathrm{Range}(\xi_q)$ is closed, by the Banach closed range theorem, $\mathrm{Null}(\xi_q)^\perp = K_V \mathrm{Range}(\xi_q)$.

Problem 2. Let $q_0 \in M$, let $C : (q, v) \in M \times V \mapsto C_q v \in Y$ linear with respect to v. Let $g : M \to \mathbb{R}$ be a function. We consider the problem of minimizing the functional:

$$J_2(q, u) = \frac{1}{2} \int_0^1 (u(t), K_{q(t)} u(t))_{X^\star, X} dt + g(q(1)) \tag{4}$$

Over all $u : [0, 1] \to X^\star$ a measurable functions and $q(\cdot) \in W_{q_0}^{1,1}([0, 1]; M)$ such that $\dot{q}(t) = K_{q(t)} u(t)$ and $C_{q(t)} K_V \xi_{q(t)}^\star u(t) = 0$ for almost every $t \in [0, 1]$.

Proposition 3. (Arguillere et al. [1]) Assume that $\mathrm{Null}(\xi_q) \subset \mathrm{Null}(C_q)$ and that $\mathrm{Range}(\xi_q)$ is closed, for every $q \in M$. Then problems 1 and 2 are equivalent. Moreover, if $(q(\cdot), u(\cdot))$ is an optimal solution of problem 2, then $(q(\cdot), v(\cdot))$ is an optimal solution of problem 1, with $v(\cdot) = K_V \xi_{q(\cdot)}^\star u(\cdot)$. Conversely, if $(q(\cdot), v(\cdot))$ is an optimal solution of problem 1 then there exists a measurable function $u : [0, 1] \to X^\star$ such that $v(\cdot) = K_V \xi_{q(\cdot)}^\star u(\cdot)$ and $(q(\cdot), u(\cdot))$ is an optimal solution of problem 2.

Example 1. *The space of landmarks*
Let $X = (\mathbb{R}^d)^n$ and $M = Lmk_d(n) := \{(x_1, \cdot, x_n) \in (\mathbb{R}^d)^n \,|\, x_i \neq x_j$ if $i \neq j\}$. With $q = (x_1, \cdot, x_n)$ the action of order $l = 0$ is defined as $\varphi \cdot q = (\varphi(x_1), \cdots, \varphi(x_n))$. And then $\xi_q(v) = v \cdot q$.
In this context, by identifying $X^\star$ and X, we can write:

$$v(t) = \sum_{i=1}^n K(x_j(t), \cdot) u_j(t), \quad K_q = [K(x_i, x_j)]_{i,j} \in \mathcal{M}_{nd}(\mathbb{R})$$

3 Constrained geodesic equations in shape spaces

3.1 First-order optimality conditions

Problem 3. Let $q_0 \in M$. Let $L : M \times V \to \mathbb{R}$, $g : M \to R$, ξ and C of class $\mathcal{C}^1$, with an additional hypothesis on L. We consider the optimal control problem of minimizing the functional:

$$J(q, v) = \int_0^1 L(q(t), v(t)) + g(q(1)) \tag{5}$$

Over all $q(\cdot) \in H^1([0, 1]; M)$ and $v(\cdot) \in L^2([0, 1]; V)$ such that $\dot{q}(t) = \xi_{q(t)}v(t)$ and $C_{q(t)}v(t) = 0$.

Definition 7. The *hamiltonian* $H : M \times X^\star \times V \times Y^\star \to \mathbb{R}$ is a function of class $\mathcal{C}^1$, defined by:

$$H(q, p, v, \lambda) = (p, \xi_q v)_{X^\star,X} - L(q, v) - (\lambda, C_q v)_{Y^\star,Y} \tag{6}$$

Theorem 2. (Arguillere et al. [1]) Assume that the linear operator $C_q : V \to Y$ is surjective for every $q \in M$. Let $q(\cdot) \in H^1_{q_0}([0, 1]; X)$ and $v(\cdot) \in L^2([0, 1]; V)$ be an optimal solution of problem 3. Then there exist $p \in H^1([0, 1]; X^\star)$ and $\lambda \in L^2([0, 1]; Y^\star)$ such that $p(1) + dgq(1) = 0$ and for almost every $t \in [0, 1]$:

$$\begin{cases} \dot{q}(t) = \partial_p H(q(t), p(t), v(t), \lambda(t)) \\ \dot{p}(t) = -\partial_q H(q(t), p(t), v(t), \lambda(t)) \\ \quad \partial_v H(q(t), p(t), v(t), \lambda(t)) = 0 \end{cases} \tag{7}$$

Proposition 4. (Arguillere et al. [1]) The differential of J_1 is given by

$$dJ_1(v)[\delta v] = -\int_0^1 \partial_v H(q_v(t), p(t), v(t))[\delta v(t)]dt \tag{8}$$

For every $\delta v \in L^2([0, 1]; V)$,where $p \in H^1([0, 1], X^\star)$ is the solution of $\dot{p}(t) = -\partial_q H(q_v(t), p(t), v(t))$ for almost every $t \in [0, 1]$ and $p(1) + dg(q_v(1)) = 0$.

3.2 The geodesic equations in a shape space

Theorem 3. (Arguillere et al. [1]) Let $q(\cdot) \in H^1_{q_0}([0, 1]; M$ and $v(\cdot) \in L^2([0, 1]; V)$ be a solution of problem 1. There exists $p \in H^1([0, 1]; X^\star)$ such that $v(t) = K_V(\xi^\star_{q(t)}p(t) - C^\star_{q(t)}\lambda(t))$, for almost every $t \in [0, 1]$, $p(1) + dgq(1) = 0$ and the geodesic equations for almost every $t \in [0, 1]$:

$$\begin{cases} \dot{q}(t) = K_{q(t)}p(t) - \xi_{q(t)}K_V C^\star_{q(t)}\lambda(t) \\ \dot{p}(t) = -\partial_q(p(t), \xi_{q(t)}v(t))_{X^\star,X} + \partial_q(\lambda(t), C_{q(t)}v(t))_{Y^\star,Y} \\ \lambda(t) = (C_{q(t)}K_V C^\star_{q(t)})^{-1}C_{q(t)}K_V\xi^\star_{q(t)}p(t) \end{cases} \tag{9}$$

4 Structured Deformation Modeling with Implicit Deformation Modules

4.1 Deformation Modules

Definition 8. We define $(M, \mathcal{H}, \xi, \zeta, c)$ as a $\mathcal{C}^k$-*deformation module* of order l with *geometrical descriptors* in M, h *controls* in $\mathcal{H}$, infinitesimal action ξ if:

The *field generator* $\zeta : M \times \mathcal{H} \to \mathcal{C}^l_0(\mathbb{R}^d, \mathbb{R}^d)$ is continuous, linear with respect to h and $\mathcal{C}^k$ with respect to q.

The *cost* c is smooth and a positive quadratic form on $\mathcal{H}$.

Example 2. *Examples of deformation modules:*
Local translations
With K_σ a Gaussian kernel, $M = \mathcal{H} = (\mathbb{R}^d)^n$, with $q = (q_1, \cdots, q_n)$, and $\varphi \cdot q = (\varphi(q_1), \cdots, \varphi(q_n))$. We can use $\zeta_q(h) = \sum_{i=1}^n K_\sigma(q_i, \cdot)h_i$ and $c_q(h) = \|\zeta_q(h)\|^2_{V_\sigma}$. And $(M, \mathcal{H}, \xi, \zeta, c)$ is a deformation module.
Local rotations/ scalings
With $H = \mathbb{R}$, $M = \mathbb{R}^2$, $\varphi \cdot q = \varphi(q)$, $\zeta_q(h) = h\sum_{i=1}^n K_\sigma(z_i(q), \cdot)d_i(q)$ and $c_q(h) = \|\zeta_q(h)\|^2_{V_\sigma}$. Then $(M, \mathcal{H}, \xi, \zeta, c)$ is a deformation module.

Definition 9. We say that a deformation module satisfies the *Uniform Embedding Condition* (UEC) if there exists a Hilbert space of vector fields V continuously embedded in $\mathcal{C}_o^{l+k}(\mathbb{R}^d, \mathbb{R}^d)$ and a constant $\gamma > 0$ such that :

$$\forall q \in M, \forall h \in \mathcal{H}, \quad \zeta_q h \in V, \quad \|\zeta_q(h)\|_V^2 \leq \gamma c_q(h) \tag{10}$$

Definition 10. Let $(M_l, \mathcal{H}_l, \xi_l, \zeta_l, c_l)$, $l = 1 \cdots L$, be L $\mathcal{C}^k$-deformation modules of order l. We define the *compound module* of modules these modules $(M, \mathcal{H}, \xi, \zeta, c)$ where $M = \prod_l M_l$, $\mathcal{H} = \prod \mathcal{H}_l$,and :

$$\begin{aligned} \zeta_q h &= \sum_l \zeta_{q_l}^l h_l \\ \xi_q v &= (\xi_{q_l}^l v_l)_l \\ c_q(h) &= \sum_l c_{q_l}^l (h_l) \end{aligned}$$

Proposition 5. (Gris et al. [3]) If $(M_l, \mathcal{H}_l, \xi_l, \zeta_l, c_l)$, with $l = 1, \cdots, L$, are $\mathcal{C}^k$-deformation modules of order l, then their compound module is also a $\mathcal{C}^k$-deformation module of order l. Furthermore, if they all satisfies UEC, then their compound module also satisfies UEC.

Definition 11. Let $(a, b) \in M^2$. We consider the problem of minimizing the functional:

$$J_{a,b}(h) = g(q(1), b) + \int_0^1 c_{q(t)}(h(t))dt \tag{11}$$

Such that $v(t) = \zeta_{q(t)} h(t)$, and with $q(t) = \varphi^v(t) \cdot a$.

Theorem 4. (Gris et al. [3]) With $(M, \mathcal{H}, \zeta, \xi, c)$ a $\mathcal{C}^k$ deformation module of order $\mathcal{C}^l$ satisfying UEC with $k, l \geq 1$. We suppose that g is $\mathcal{C}^1$. If $h \in L^2([0,1], \mathcal{H})$ minimizes functional $J_{a,b}$ then there exists a path $p : t \in [0,1] \to p(t) \in T^\star_{q(t)} M$ such that with :

$$H(q, p, h) = (p, \xi_q \zeta_q h) - \frac{1}{2} c_q(h) \tag{12}$$

The *hamiltonian* of the system and $p(1) + \partial_q g(q(1), b) = 0$:

$$\begin{cases} \dot{q}(t) = \xi_{q(t)} \zeta_{q(t)} h(t) \\ \dot{p}(t) = -\dfrac{\partial H}{\partial q}(q(t), p(t), h(t)) \\ \quad 0 = \dfrac{\partial H}{\partial h}(q(t), p(t), h(t)) \end{cases} \tag{13}$$

4.2 Implicit Deformation Modules

Definition 12. Let M be a general -shape space of order $l \geq 1$ (the space of geometrical descriptors), $\mathcal{H}$ be a vector space of finite dimension (the space of controls) and V be a RKHS continuously embedded in $\mathcal{C}_0^l(\mathbb{R}^d)$ with $l \geq 1$. We consider
A *space of constraints* Y which is a vector space of finite dimension.
An *evaluation operator* $S : M \times V \to Y$, linear in v, globally smooth such that for all $q \in M, S(q, \cdot)$ is surjective.
A *prescription operator* $A : M \times \mathcal{H} \to Y$, linear in $h \in \mathcal{H}$ and smooth.
And a *relaxing constant* $\nu > 0$.

Proposition 6. (Charlier et al. [2])
Let $(M, \mathcal{H}, \xi, \zeta, c)$ be the *implicit deformation module* induced by the constraints (S, A). With $K : V^\star \to V$ the kernel of the RKHS V. For all $(q, h) \in M \times \mathcal{H}$, there exists a unique vector field $\zeta_q(h)$ in V satisfying $\zeta_q(h) = \underset{v \in V}{\text{Argmin}} \frac{1}{\nu} \|S_q(v) - A_q(h)\|^2 + \|v\|_V^2$
It is defined by $\zeta_q(h) = K S_q^\star \lambda$ with $\lambda = (S_q K S_q^\star + \nu I)^{-1} A_q(h) \in Y^\star$
The associated cost is then $c_q(h) = (\lambda | A_q(h))$ and $(M, \mathcal{H}, \xi, \zeta, c)$ is an (explicit) deformation module in the sense of definition 8.

References

[1] Sylvain Arguillere et al. "Shape deformation analysis from the optimal control viewpoint". In: *Journal de mathématiques pures et appliquées* 104.1 (2015), pp. 139–178.

[2] Benjamin Charlier et al. "Structured Deformation Modeling with Implicit Deformation Modules". In: (2026).

[3] Barbara Gris, Stanley Durrleman, and Alain Trouvé. "A sub-riemannian modular framework for diffeomorphism-based analysis of shape ensembles". In: *SIAM Journal on Imaging Sciences* 11.1 (2018), pp. 802–833.

A Proof notes

Proof. Structure of the proof of proposition 1, from [1]

Using Faà di Bruno's formula, with Π_n the set of partitions of $\{1, \cdots, n\}$ ($\forall \pi \in \Pi_n$, $\pi = \{B_1, \cdots B_k\}$,with $k = |\pi|$, such that $\bigsqcup_{i=1}^{k} B_i = \{1, \cdots n\}$).

$$D^n(f \circ g)(x) = \sum_{\pi \in \Pi_n} D^{|\pi|} f(g(x))[D^{|B_1|} g(x), \cdots, D^{|B_k|} g(x)]$$

Proof of this formula by induction using

$$D\{D^{|\pi|} f(g(\cdot))[D^{|B_1|} g(\cdot), \cdots, D^{|B_k|} g(\cdot)]\}(x) =$$
$$\sum_{i=1}^{k} D^{|\pi|} f(g(x))[D^{|B_1|} g(x), \cdots, D^{|B_i|+1} g(x), \cdots, D^{|B_k|} g(x)] + D^{|\pi|+1} f(g(x))[D^{|B_1|} g(x), \cdots, D^{|B_k|} g(x), D^1 g(x)]$$

Thus, with $t \in [0, 1]$ fixed, applying this formula to $f = v(t)$ and $g = \varphi(t)$, and with $\frac{d}{dt} D^n \varphi(t) = D^n \frac{d}{dt} \varphi(t) = D^n f \circ g$ we have using $k = 1 \Leftrightarrow \pi = \{\{1, \cdots, n\}\}$ and $k > 1 \Leftrightarrow \forall i \, |B_i| \leq n - 1$

$$\begin{aligned} \|D^n f \circ g\| &\leq \sum_{\pi \in \Pi_n} \|D^{|\pi|} f(g(x))\| \prod_{i=1}^{k} \|D^{|B_i|} g(x)\| \leq \sum_{\pi \in \Pi_n} \|f\|_{|\pi|} \prod_{i=1}^{k} \|g\|_{|B_i|} \\ &\leq \|f\|_1 \|\|g\|_n + \|f\|_n P_i(\|g\|_1, \cdots \|g\|_{n-1}) \end{aligned}$$

With P_i universal multivariate polynomials depending only of n. Using the fact that with $u > 0$, $\{r | \, \exists a > 0, \, r \leq a \exp(au)\}$ is closed under multinomial combinations and using Grönwall applied to $t \to \|D^n \varphi(t)\|$ yields the following inequality with universals constants D_n, using a strong induction over $i \leq n$:

$$\|\varphi(t) - \mathrm{Id}_{\mathbb{R}^d}\|_n \leq D_n \exp\left(D_n \int_0^1 \|v(s)\|_n ds\right) \tag{14}$$

□

Proof. Structure of the proof of proposition 2, from [1] The proof is based the definition of ξ, i.e. $(I_d + \delta\varphi) \cdot q_0 = q_0 + \xi_{q_0} \delta\varphi + o(\delta\varphi)$, using $o(\delta\varphi \circ \varphi^{-1}) = o(\delta\varphi)$ and on the equation:

$$(\varphi + \delta\varphi) \cdot q = (I_d + \delta\varphi \circ \varphi^{-1}) \cdot (\varphi \cdot q) = \varphi \cdot q + \xi_{\varphi \cdot q} \delta\varphi \circ \varphi^{-1} + o(\delta\varphi)$$

□

Proof. Structure of the proof of theorem 1, from [1]

We start by defining a minimising sequence $(v_n)_{n \in \mathbb{N}}$ of problem 1 , such that $v_n \in L^2([0, 1], V)$, and $q_n = \varphi_n \cdot q_0$.

This sequence is bounded because g is bounded below. We can extract $v_{n_k} \rightharpoonup \bar{v} \in L^2$ by James theorem (reflexivity) on L^2. For any compact $U \subset \mathbb{R}^d$, we can apply Arzelà–Ascoli theorem to (φ_n) restricted to U, because :

Indeed $\varphi_{n|U}$ is uniformly bounded using eq. (2) $\|\varphi_n(t) - \mathrm{Id}_{\mathbb{R}^d}\|_l \leq C \exp\left(C \sup_{n\in\mathbb{N}} \|v_n\|_{L^2_V}\right)$.

And $\varphi_{n|U}$ is equicontinuous, $\|\varphi_n(t) - \varphi_n(t')\|_l \leq \sqrt{|t-t'|} C \exp\left(C \sup_{n\in\mathbb{N}} \|v_n\|_{L^2_V}\right)$.

Then this sequence is on a compact of $\mathcal{C}_0([0,1], \mathcal{C}^l_0(U, \mathbb{R}^d))$. By a diagonal extraction we can defined an extraction converging to $\bar{\varphi}$ on any compact of $\mathbb{R}^d$.

And, using the weak convergence of v_{n_k} in L^2_V and the uniform convergence of φ_n on any compact, we can prove that $\dot{\bar{\varphi}}(t) = \bar{v}(t) \circ \bar{\varphi}(t)$:

We can define $\chi_{[0,t]}\delta_f \in V^\star$ by $(\chi_{[0,t]}\delta_f, v) = \int_0^t v \circ f(s) ds$. Then using $\{x\}$ as a compact.

$$|\varphi_{n_k}(t,x) - x - \int_0^t \bar{v} \circ \bar{\varphi}(s,x) ds| \leq \left(\sup_n \int_0^1 \|v_n(t)\|_1\right) \sup_{s\in[0,t]} |\varphi_{n_k}(s,x) - \bar{\varphi}(s,x)| + |(\chi_{[0,t]}\delta_{\bar{\varphi}(\cdot,x)} | v_{n_k} - \bar{v})|$$

Then we use that q_0 as compact support to prove the convergence of q_n toward $\bar{q} = \bar{\varphi} \cdot q_0$. We can verify by continuity that $C_{\bar{q}}\bar{v} = 0$.

Finally using the lower semi-continuity of J_1, and the weak convergence, we obtained $J_1(\bar{q}, \bar{v}) \leq \lim_j J_1(q_{n_j}, v_{n_j}) = \inf J_1$. □

Proof. Structure of the proof of proposition 3, from [1]

For any $u : [0,1] \to X^\star$, we can define $v(t) = K_V \xi_{q(t)} u(t)$ which belong to $L^2([0,1], V)$ and then $\dot{q}(t) = K_{q(t)} u(t) = \xi_{q(t)} v(t)$, thus $\inf J_1 \leq \inf J_2$.

For any $\epsilon > 0$, let $v \in L^2([0,1], V)$ and $q \in H^1_{q_0}([0,1], M)$ such that $J_1(q,v) \leq \inf J_1 + \epsilon$. We can write $v(t) = v_1(t) + v_2(t)$ under the decomposition $\mathrm{Null}(\xi_{q(t)}) \oplus \mathrm{Null}(\xi_{q(t)})^\perp$. But $\mathrm{Null}(\xi_{q(t)})^\perp = \mathrm{Range}(K_V \xi^\star_{q(t)})$ thus $v_2(t) = K_V \xi^\star_{q(t)} u_2(t)$. Moreover $\|v_2(t)\|^2_V \leq \|v(t)\|^2_V$ thus $\inf J_2 \leq J_2(u_2) = J_1(v_2) \leq J_1(v) \leq \inf J_1 + \epsilon$. □

Proof. Structure of the proof of proposition 5, from [3]

First, it is a $\mathcal{C}^k$-deformation module of order l because the field generator $\zeta : q, h \in M \times \mathcal{H} \mapsto \zeta_q h = \sum_l \zeta^l_{q_l} h_l \in \mathcal{C}^l_0(\mathbb{R}^d, \mathbb{R}^d)$ is continuous, linear with respect to h and $\mathcal{C}^k$ with respect to q. And the *cost* $c : q, h \mapsto c_q(h) = \sum_l c^l_{q_l}(h_l)$ is smooth and a positive quadratic form on $\mathcal{H}$, where $M = \prod_l M_l$, $\mathcal{H} = \prod \mathcal{H}_l$. And the infinitesimal action is given by $\xi_q v = (\xi^l_{q_l} v_l)_l$.

Then, we need to prove that this deformation module satisfy the UEC. We define $\sigma : (w_1, \cdots, w_L) \in W = \prod_l V_l \mapsto \sum_l w_l$. Then with $V = \sigma(W)$ equipped with the norm $|v|^2_V = \inf\{\sum_l |v_l|^2_{V_l} \,|\, \sigma((v_l)_l) = v\}$. And

$$|\zeta_q(h)|^2_V \leq \sum_l |\zeta^l_{q_l} h_l|^2_{V_l} \underbrace{\leq}_{UECl} \sum_l \gamma_l c^l_{q_l}(h_l) \leq (\max_l \gamma_l)\, c_q(h)$$

□

B Algorithms

Algorithm 1: Hamiltonian Deformation under constraint using forward Euler

```
procedure Hamiltonian Deformation
    Input q_0, p_0, δ_t: q_0 : initial point cloud, p_0: initial momentum
    q, p ← q_0, p_0
    for t ∈ {0, δ_t, ··· , 1 − δ_t} do
        v ← K_V(ξ*_q p − C*_q λ)
        λ ← (C_{q(t)} K_V C*_{q(t)})^{−1} C_{q(t)} K_V ξ*_{q(t)} p(t)
        q̇ ← K_q p − ξ_q K_V C*_q λ
        ṗ ← −∂_q(p(t), ξ_q v) + ∂_q(λ, C_q v)
        q, p ← q + δ_t q̇, p + δ_t ṗ # Forward Euler.
```